

# Clean Fixed Circles Obstruct Isolated-Orbit Determinants on a Heisenberg Nilmanifold

Route-A structural certificate C146

## Abstract

We freeze an explicit integer-lattice automorphism of the three-dimensional Heisenberg group. Its horizontal action is hyperbolic, while its center is fixed pointwise. Every iterate therefore owns a clean central fixed circle, has singular ordinary isolated-orbit stability factor, and has zero Lefschetz number. The horizontal torus provides an exact isolated-point control through iterate twenty. These are source-side obstruction results, not a target or arithmetic construction.

## 1 The frozen lattice automorphism

Write the real Heisenberg group as  $H = \mathbb{R}^3$  with

$$(x, y, z)(X, Y, Z) = (x + X, y + Y, z + Z + xY), \quad \Gamma = \mathbb{Z}^3.$$

On the compact left quotient  $N = \Gamma \backslash H$ , freeze

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad q(x, y) = x(x - 1) + xy + \frac{y(y - 1)}{2},$$

and

$$\Phi(x, y, z) = (2x + y, x + y, z + q(x, y)). \quad (1)$$

The central correction is forced by the coordinate cocycle. Direct expansion gives

$$q(v + w) - q(v) - q(w) = 2xX + xY + Xy + yY,$$

which equals  $(2x + y)(X + Y) - xY$ . Thus (1) is a group homomorphism. Moreover,  $q(m, n)$  is integral for integer  $m, n$ , so the map preserves  $\Gamma$ . Since  $A^{-1}$  is integral and the inverse central correction is  $-q(A^{-1}v)$ , the inverse preserves  $\Gamma$  as well. Hence (1) is a lattice automorphism and descends to  $N$ .

## 2 Clean fixation and singular stability

The embedded central circle

$$C = \{[0, 0, z] : z \in \mathbb{R}/\mathbb{Z}\}$$

is fixed pointwise by  $\Phi$ , hence by every positive iterate. The horizontal eigenvalues are  $(3 \pm \sqrt{5})/2$ , neither of whose positive powers is one. Consequently the horizontal fixed classes at each iterate are discrete, and  $C$  is a one-dimensional clean fixed component.

The derivative is block lower triangular. For every  $n \geq 1$ , its diagonal blocks are  $A^n$  and 1, so

$$\det(I - D\Phi^n) = \det(I - A^n)(1 - 1) = 0. \quad (2)$$

Thus the usual stability denominator for an isolated periodic orbit is singular at every period. We do not introduce a clean-family regularization.

Let  $\alpha = dx$ ,  $\beta = dy$ , and  $\gamma = dz - x dy$ , with  $d\gamma = -\alpha \wedge \beta$ . Heisenberg cohomology has dimensions  $(1, 2, 2, 1)$ . Pullback has trace  $\text{tr}(A^n)$  in both middle degrees and acts as the identity in degrees zero and three. Therefore

$$L(\Phi^n) = 1 - \text{tr}(A^n) + \text{tr}(A^n) - 1 = 0. \quad (3)$$

### 3 Horizontal control and exact validation

On the horizontal torus, fixed classes form  $\ker(A^n - I : \mathbb{T}^2 \rightarrow \mathbb{T}^2)$ , so their number is

$$|\det(A^n - I)| = \operatorname{tr}(A^n) - 2 = L_{2n} - 2, \quad (4)$$

where  $L_k$  is the Lucas sequence. Here  $A = Q^2$  for  $Q = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ , proving the last equality. Exact evidence records (4), (2), and (3) through  $n = 20$ .

The toral number in (4) is not a nilmanifold component count. At  $n = 2$  the horizontal class  $v = (1/5, 2/5)$  satisfies  $(A^2 - I)v = (2, 1)$ , but  $q(v) + q(Av) = 0$ . If a point  $(v, z)$  were fixed on the left quotient, some  $(2, 1, k) \in \Gamma$  would obey  $\Phi^2(v, z) = (2, 1, k)(v, z)$ . Its central coordinate would require  $0 = k + 2(2/5)$ , impossible for integral  $k$ . Thus this base class does not lift to a fixed circle, and we assert no full component count for  $\operatorname{Fix}(\Phi^n)$ .

The independent checker passes 687 assertions, the SymPy reconstruction passes 87 checks, byte replay passes, and all 31 hostile receipts (30 repaired- hash semantic mutations and one stale hash) are rejected.

### 4 Route-A boundary

The strict tuple is (A1\_FAIL, A2\_FAIL, A3\_FAIL, A4\_FORMAL\_HINT), overall ROUTE\_A\_EXPLORATORY. Haar preservation gives a natural Koopman unitary  $U_\Phi f = f \circ \Phi$  on  $L^2(N)$  with the same iterate clock, but no bridge to the singular clean-family weights; it is only a formal hint. We claim no isolated primitive-orbit ledger, target divisor, target functional equation or counting law, arithmetic/local factor, Euler factor, root number, automorphy, Hilbert–Pólya construction, or Route-B authorization. The scope literal is NO\_BAD\_EULER\_OR\_ROOT\_NUMBER.